Pavel Vasilyev

Senior AI Engineer - LLM Systems, APIs & Applied ML

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PROFILE

Software engineer with a research and applied-mathematics background. Fifteen years of programming, analytics, and systems design, then four years of vibe coding with LLMs.

SKILLS

Foundations: Programming, analytics, systems design, architecture since 2009; research, applied math.

LLM systems: Tool calling, provider routing and fallback, quota and rate-limit handling, cost and latency tracing. Claude, Gemini, OpenRouter, Ollama, Cloudflare Workers AI.

Mathematics and ML: Topological data analysis, quantitative modelling, NLP, feature engineering, model training.

Languages: Rust, Elixir, Python, Swift, Clojure, TypeScript; earlier Haskell and Agda.

Infrastructure: Nix, Caddy, systemd, CI, PWA build and deploy tooling.

EXPERIENCE

Independent Software Engineer - self-employed 2010-present

LLM tooling, PWAs, Rust, Elixir

Design, build, and ship software for clients and my own projects: 150+ projects across eight or more languages, with LLM coding agents as my main tool since 2022.

The recent work targets what makes AI-assisted development safe to rely on: provider routing and fallback, quota handling, and deploy pipelines that check themselves.

Research Lead - Spring Research 2020-2022

Quant modelling, feature engineering

Owned applied research on financial-market data: signal representations, features, model selection, training.

I built and trained the models and carried findings from exploratory analysis into the team's shared modelling.

Earlier Data Science, NLP & Software Roles - Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, and Finfon 2009-2019

NLP, classification, computer vision

Financial-news signal extraction, medical-text classification, vision pipelines, corpus infrastructure; Tel Aviv, Hong Kong, Prague.

SELECTED WORK

pocket - Suite of 18 installable PWAs behind one pipeline: machine-checked rules (offline cache, update banner, safe-area, no critical-path network), content-hash service-worker versions, and a deploy that verifies the live URLs from outside. Live; code private. **→ End-to-end delivery, from build to verified deploy.**

LLM routing and quota tooling - Elixir, Python, and Rust: provider routing and fallback chains, jittered retry honoring Retry-After, per-client rate limits, token, cost, and latency audit logs. Python harness over 88 requests: 100% success on a quota probe, 83% in an outage probe. Private. **→ Cost, latency, and quota visible per call.**

topo-sonify - Rust library that maps persistent homology, Hodge theory, and Morse theory to sound. The math core was checked against known invariants ($\beta(S^2) = (1,0,1)$, $\chi = 2$) before any audio code; unit-tested, CI green. Public. **→ Applied-math depth behind the engineering.**

EDUCATION

Education: MSc Computational Linguistics, RSUH, Moscow, 2015. BSc Nuclear Physics, Czech Technical University, Prague, 2009.

Languages: English, Hebrew, Russian; working knowledge of German and French.